Data Communication & Computer Networks — Topic 2: Network Models (OSI & TCP/IP)

Practice Test Booklet — 100 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/ DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — OSI Model: Basics, Layer Order & Support Layers

Q1. The OSI model was developed by:

OSI model किसने develop किया?

(A) IEEE **(B)** ISO
(C) IETF **(D)** ITU

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q2. OSI stands for:

OSI का पूरा नाम (full form) क्या है?

(A) Open Systems Interconnection **(B)** Operating System Interface
(C) Optical Service Implementation **(D)** None of these

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q3. The OSI model has how many layers?

OSI model में कितनी layers हैं?

(A) 5 **(B)** 6
(C) 7 **(D)** 8

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q4. Why was the OSI model developed?

OSI model क्यों develop किया गया?

(A) Manufacturers disliked the TCP/IP protocol suite / Manufacturers को TCP/IP protocol suite पसंद नहीं था **(B)** The rate of data transfer was increasing exponentially / Data transfer की दर तेज़ी से बढ़ रही थी
(C) Standards were needed to allow any two systems to communicate / किन्हीं भी दो systems के बीच communication के लिए standards चाहिए थे **(D)** None of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q5. In the OSI model, when data is sent from device A to device B, which is the 5th layer to receive data at B (counting from the bottom)?

OSI model में जब data device A से device B को भेजा जाता है, तो B पर data पाने वाली 5वीं layer (नीचे से गिनती में) कौन-सी है?

(A) Application layer **(B)** Transport layer
(C) Link layer **(D)** Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q6. Network support layers are:

Network support layers कौन-सी हैं?

(A) 1, 2, 3 **(B)** 4, 5
(C) 5, 6, 7 **(D)** 4 only

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q7. Which layer connects the network support layers and the user support layers?

कौन-सी layer network support layers और user support layers को जोड़ती है?

(A) Transport layer **(B)** Network layer
(C) Data Link layer **(D)** Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q8. In the ISO-OSI reference model, the Session layer offers dialog control, token management and _____ as services.

ISO-OSI reference model में Session layer dialog control, token management और _____ services देती है।

(A) Synchronization **(B)** Asynchronization
(C) Errors **(D)** Flow control

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q9. How many address types (identifiers) are used from sender to receiver in OSI/TCP-IP communication?

OSI/TCP-IP communication में sender से receiver तक कितने address types (identifiers) use होते हैं?

(A) 2 **(B)** 3
(C) 4 **(D)** 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q10. Who developed the standards for the OSI reference model?

OSI reference model के standards किसने develop किए?

(A) ANSI - American National Standards Institute **(B)** ISO - International Organization for Standardization
(C) IEEE - Institute of Electrical and Electronics Engineers **(D)** ACM - Association for Computing Machinery

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q11. In the OSI model, layers 5, 6 and 7 are mainly called:

OSI model में layers 5, 6, 7 को मुख्यतः कहा जाता है:

(A) Support layers **(B)** User support layers
(C) Network layers **(D)** Physical layers

[Source: Practice — original (textbook-based)] · Medium

Q12. Two communicating hosts exchange data between the same layers using:

दो hosts एक ही layers के बीच data किससे बदलते हैं?

(A) Interfaces **(B)** Protocols (peer-to-peer)
(C) Cables **(D)** Ports

[Source: Practice — original (textbook-based)] · Medium

Q13. In the OSI model, error control as well as flow control are performed by which TWO layers?

OSI model में error control और flow control दोनों किन दो layers में होते हैं?

(A) Physical and Network / Physical और Network **(B)** Data Link and Transport / Data Link और Transport
(C) Transport and Session / Transport और Session **(D)** Network and Session / Network और Session

[Source: Practice — original (textbook-based)] · Medium

Q14. The session, presentation, and application layers are the _____ support layers.

Session, Presentation और Application layers _____ support layers हैं।

(A) user **(B)** network
(C) both (a) and (b) **(D)** neither (a) nor (b)

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

Q15. Which of the following is NOT a network support layer?

निम्नलिखित में से कौन network support layer नहीं है?

(A) Transport Layer **(B)** Network Layer
(C) Data Link Layer **(D)** Physical Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

Section B — OSI Layer Functions (Layer 1 → Layer 7)

Q16. Fiber optic cables and coaxial cables function at which OSI layer?

Fiber optic cables और coaxial cables किस OSI layer पर काम करती हैं?

(A) Data Link Layer **(B)** Physical Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q17. The Physical layer deals with:

Physical layer किससे निपटती है?

(A) Logical addressing **(B)** Media, signals and binary transmission / Media, signals और binary transmission
(C) Error correction **(D)** Segmentation

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q18. The sub-layer of the Physical layer which directly interfaces with the MAC sub-layer is called:

Physical layer का वह sub-layer जो सीधे MAC sub-layer से जुड़ता है, कहलाता है:

(A) Physical signalling sub-layer **(B)** Physical data sub-layer
(C) Physical address sub-layer **(D)** Physical transport sub-layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q19. Which layer creates the 'collision domain'?

'Collision domain' कौन-सी layer बनाती है?

(A) Physical Layer (hubs) **(B)** Data Link Layer (switches/bridges)
(C) Network Layer **(D)** Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q20. Data Link Layer is divided into:

Data Link Layer किनमें विभाजित है?

(A) LLC and MAC **(B)** Physical and Network
(C) Transport and Session **(D)** Application and Presentation

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q21. Physical (hardware) addressing is added at:

Physical (hardware) addressing किस layer पर जोड़ी जाती है?

(A) Network **(B)** Transport
(C) Data Link **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q22. 'Transparent transmission' is the function of which layer?

'Transparent transmission' किस layer का काम है?

(A) Data link **(B)** Network
(C) Session **(D)** Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q23. Which layer handles 'hop-to-hop' flow control?

'Hop-to-hop' flow control कौन-सी layer संभालती है?

(A) Network **(B)** Data Link
(C) Transport **(D)** Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q24. Which OSI layer is responsible for logical addressing?

Logical addressing के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Data Link **(B)** Network
(C) Transport **(D)** Application

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q25. Which layer performs segmentation?

Segmentation कौन-सी layer करती है?

(A) Data link **(B)** Transport
(C) Network **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q26. Which layer provides end-to-end communication?

End-to-end communication कौन-सी layer देती है?

(A) Network **(B)** Transport
(C) Session **(D)** Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q27. In the OSI model, which layer(s) participate in 'end-to-end' delivery? (1) Network (2) Transport (3) Session (4) Data link

OSI model में 'end-to-end' delivery में कौन-सी layer(s) भाग लेती हैं? (1) Network (2) Transport (3) Session (4) Data link

(A) 1, 2 only **(B)** 2 only
(C) 1, 2, 3 **(D)** 2, 3, 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q28. Which layer performs 'service point addressing'?

'Service point addressing' कौन-सी layer करती है?

(A) Network **(B)** Data Link
(C) Transport **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q29. Which OSI layer handles message boundaries (knowing where one message ends and the next begins)?

कौन-सी OSI layer message boundaries संभालती है (एक message कहाँ समाप्त होता है और अगला कहाँ शुरू)?

(A) Presentation **(B)** Session
(C) Transport **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q30. Determine the OSI layer responsible for dialogue control (half-duplex or full-duplex communication).

Dialogue control (half-duplex या full-duplex communication) के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Session Layer **(B)** Transport Layer
(C) Network Layer **(D)** Data Link Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q31. Which layer supports 'checkpointing' (synchronisation for recovery)?

कौन-सी layer 'checkpointing' (recovery के लिए synchronization) support करती है?

(A) Transport **(B)** Application
(C) Session **(D)** Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q32. Which layer translates data from a format used by the application into a common format (e.g., ASCII to EBCDIC)?

कौन-सी layer data को application के format से common format में बदलती है (जैसे ASCII से EBCDIC)?

(A) Application Layer **(B)** Presentation Layer
(C) Session Layer **(D)** Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q33. Which layer is responsible for encryption?

Encryption के लिए कौन-सी layer जिम्मेदार है?

(A) Session **(B)** Presentation
(C) Application **(D)** Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q34. Which OSI layer ensures interoperability between different data formats?

कौन-सी OSI layer अलग-अलग data formats के बीच interoperability सुनिश्चित करती है?

(A) Presentation **(B)** Application
(C) Session **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q35. Which OSI layer decides the type of compression?

कौन-सी OSI layer compression का प्रकार तय करती है?

(A) Application **(B)** Presentation
(C) Transport **(D)** Data link

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q36. The term 'Baud Rate' is associated with which layer?

'Baud Rate' शब्द किस layer से जुड़ा है?

(A) Network Layer **(B)** Data Link Layer
(C) Physical Layer **(D)** Application Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q37. The layer used to deal with mechanical and electrical specifications is:

Mechanical और electrical specifications से निपटने वाली layer कौन-सी है?

(A) Physical Layer **(B)** Data Link Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q38. The Physical layer is responsible for which of the following operations?

Physical layer निम्नलिखित में से किन operations के लिए जिम्मेदार है?

(A) Line coding **(B)** Channel coding
(C) Modulation **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q39. Transmission data rate is decided by:

Transmission data rate कौन तय करती है?

(A) Network layer **(B)** Physical layer
(C) Data Link layer **(D)** Transport layer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

Q40. The physical layer translates logical communication requests from the _____ into hardware-specific operations.

Physical layer, _____ के logical communication requests को hardware-specific operations में बदलता है।

(A) Network layer **(B)** Transport layer
(C) Data link layer **(D)** Application layer

[Source: InstrumentationTools 150 CN MCQs (key corrected vs Forouzan)] · Medium

Q41. The Application layer is implemented in:

Application layer कहाँ implement होती है?

(A) End system **(B)** NIC
(C) Ethernet **(D)** Packet transport

[Source: Sanfoundry CN MCQs — Reference Models] · Medium

Q42. Which statement best describes the difference between the LLC and MAC sub-layers?

LLC और MAC sub-layers के बीच अंतर सबसे अच्छा कौन बताता है?

(A) LLC handles physical addressing, MAC handles flow control / LLC physical addressing करता है, MAC flow control करता है **(B)** LLC provides the service interface to the Network layer; MAC controls access to the medium / LLC, Network layer को service interface देता है; MAC medium तक access control करता है
(C) LLC works only in wireless networks, MAC only in wired networks / LLC सिर्फ wireless में, MAC सिर्फ wired में काम करता है **(D)** LLC performs framing, MAC performs routing / LLC framing करता है, MAC routing करता है

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

Q43. A frame is received at the Data Link layer with an incorrect FCS. What will happen?

Data Link layer पर frame गलत FCS के साथ प्राप्त होता है। क्या होगा?

(A) Frame discarded **(B)** Frame corrected
(C) Sent to network layer **(D)** Re-transmitted automatically

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q44. The _____ layer is responsible for delivering data units from one station to the next without errors.

_____ layer data units को एक station से अगले station तक बिना errors पहुँचाने के लिए जिम्मेदार है।

(A) Transport **(B)** Network
(C) Data link **(D)** Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q45. _____ Layer is responsible for framing, error detection and physical addressing.

_____ Layer framing, error detection और physical addressing के लिए जिम्मेदार है।

(A) Network **(B)** Transport
(C) Application **(D)** Data Link

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q46. From which layer does the Data Link layer collect packets and encapsulate them into frames for transmission?

Data Link layer किस layer से packets लेकर उन्हें transmission के लिए frames में encapsulate करती है?

(A) Transport layer **(B)** Application layer
(C) Network layer **(D)** Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q47. IPv4 and IPv6 work at which OSI layer?

IPv4 और IPv6 किस OSI layer पर काम करते हैं?

(A) Data link layer **(B)** Network layer
(C) Transport layer **(D)** Application layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q48. The _____ layer is responsible for the source-to-destination delivery of a packet across multiple network links.

_____ layer एक packet की multiple network links के बीच source-to-destination delivery के लिए जिम्मेदार है।

(A) Transport **(B)** Network
(C) Data link **(D)** Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q49. Which layer is responsible for reliable delivery in the OSI model?

OSI model में reliable delivery के लिए कौन-सी layer जिम्मेदार है?

(A) Session **(B)** Transport
(C) Network **(D)** Application

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q50. Which layer is responsible for process-to-process delivery?

Process-to-process delivery के लिए कौन-सी layer जिम्मेदार है?

(A) Network layer **(B)** Transport layer
(C) Session layer **(D)** Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q51. Which OSI layer is responsible for encryption and compression?

Encryption और compression के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Presentation Layer **(B)** Session Layer
(C) Transport Layer **(D)** Network Layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q52. Which layer helps to present data in one form on a host to another host in its native representation?

कौन-सी layer एक host पर किसी form में data को दूसरे host पर उसकी native representation में प्रस्तुत करने में मदद करती है?

(A) Application Layer **(B)** Presentation Layer
(C) Session Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q53. _____ layer defines how data is formatted, presented, encoded, and converted for use on the network.

_____ layer तय करती है कि data network में उपयोग के लिए कैसे format, present, encode और convert होता है।

(A) Transport **(B)** Presentation
(C) Application **(D)** Physical

[Source: Central exams — SSC/Bank/Insurance Computer Awareness (BYJU'S)] · Medium

Q54. Which layer provides services to the user?

कौन-सी layer user को services देती है?

(A) Application layer **(B)** Session layer
(C) Presentation layer **(D)** Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q55. A server at a company is using TCP/IP to connect to the network and is running multiple applications. At which OSI layer is this server operating?

एक company का server network से जुड़ने के लिए TCP/IP use कर रहा है और अनेक applications चला रहा है। यह server किस OSI layer पर काम कर रहा है?

(A) Application Layer **(B)** Physical Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q56. Simple Network Management Protocol (SNMP) runs on:

Simple Network Management Protocol (SNMP) किस layer पर चलता है?

(A) Application layer **(B)** Transport layer
(C) Network layer **(D)** Data link layer

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q57. Consider the statements: S1: Network layer handles both logical addressing and routing. S2: Transport layer handles segmentation and reassembly. Which is correct?

Statements देखिए: S1: Network layer logical addressing और routing दोनों संभालती है। S2: Transport layer segmentation और reassembly संभालती है। कौन सा सही है?

(A) Only S1 **(B)** Only S2
(C) Both S1 and S2 **(D)** Neither S1 nor S2

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q58. Which layer adds a trailer (FCS) to the frame?

कौन-सी layer frame में trailer (FCS) जोड़ती है?

(A) Physical **(B)** Data Link
(C) Network **(D)** Transport

[Source: Practice — original (textbook-based)] · Medium

Q59. A router operates at which layer of the OSI model?

Router OSI model की किस layer पर काम करता है?

(A) Data Link **(B)** Network
(C) Transport **(D)** Session

[Source: Practice — original (textbook-based)] · Medium

Q60. Encryption and compression of data are the responsibilities of:

Data का encryption और compression किसकी ज़िम्मेदारी है?

(A) Transport layer **(B)** Presentation layer
(C) Session layer **(D)** Application layer

[Source: Practice — original (textbook-based)] · Medium

Q61. Establishing, managing and terminating sessions is done by:

Sessions की स्थापना, प्रबंधन और समाप्ति किसका काम है?

(A) Session layer **(B)** Transport layer
(C) Network layer **(D)** Data Link layer

[Source: Practice — original (textbook-based)] · Medium

Q62. The PDU of the network layer is called:

Network layer की PDU कहलाती है:

(A) Frame **(B)** Segment
(C) Packet **(D)** Bits

[Source: Practice — original (textbook-based)] · Medium

Q63. Converting the outgoing data into signals is the job of:

जाते data को signals में बदलना किसका काम है?

(A) Application layer **(B)** Network layer
(C) Physical layer **(D)** Session layer

[Source: Practice — original (textbook-based)] · Medium

Q64. Dialog control and synchronization of data exchange between two applications is the responsibility of which OSI layer?

दो applications के बीच dialog control और data exchange का synchronization किस OSI layer का काम है?

(A) Session layer **(B)** Transport layer
(C) Presentation layer **(D)** Application layer

[Source: Practice — original (textbook-based)] · Medium

Q65. Repeaters function in:

Repeaters किस layer पर काम करते हैं?

(A) Physical layer **(B)** Data link layer
(C) Network layer **(D)** Both (A) and (B)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q66. The physical layer is responsible for:

Physical layer किसके लिए जिम्मेदार है?

(A) Bit-by-bit delivery **(B)** Port to port delivery
(C) Process to process delivery **(D)** Application to application delivery

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q67. Which one of the following is a data link (Layer 2) protocol?

निम्न में से कौन data link (Layer 2) protocol है?

(A) Point-to-Point Protocol (PPP) **(B)** HDLC
(C) Ethernet **(D)** All of these

[Source: InstrumentationTools 150 CN MCQs] · Hard

Q68. Which layer of the OSI model is responsible for the transmission and reception of unstructured raw data between a device and a physical transmission medium?

OSI model की कौन-सी layer device और physical transmission medium के बीच unstructured raw data के transmission और reception के लिए जिम्मेदार है?

(A) Transport layer **(B)** Data-Link layer
(C) Network layer **(D)** Physical layer

[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)] · Hard

Q69. Bridge works in which layer of the OSI model?

Bridge OSI model की किस layer पर काम करता है?

(A) Application layer **(B)** Transport layer
(C) Network layer **(D)** Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2) · also asked in KVS PGT CS 2018 Official Paper] · Hard

Q70. The reliable data transfer is implemented by:

Reliable data transfer कौन-सी layer implement करती है?

(A) Network layer **(B)** Physical layer
(C) Presentation layer **(D)** Transport layer

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q71. If a port number is 16 bits (2 bytes), what is the MINIMUM size of the transport-layer header needed to carry the port addresses?

यदि port number 16 bits (2 bytes) का है, तो port addresses ले जाने के लिए transport-layer header की MINIMUM size कितनी होगी?

(A) 2 bytes **(B)** 4 bytes
(C) 8 bytes **(D)** 20 bytes

[Source: Forouzan Ch1 (Q1-23) — Practice Set] · Hard

Q72. Application layer protocol defines:

Application layer protocol क्या define करता है?

(A) Types of messages exchanged / कौन-से messages आदान-प्रदान होते हैं **(B)** Rules for when and how processes send and respond to messages / processes messages कब और कैसे भेजें/जवाब दें — इसके rules

(C) Message format, syntax and semantics / Message format, syntax और semantics **(D)** All of the above / उपरोक्त सभी

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

Q73. SMTP, FTP and DNS are the protocols of which layer of the OSI model?

SMTP, FTP और DNS OSI model की किस layer के protocols हैं?

(A) Network layer **(B)** Transport layer

(C) Application layer **(D)** Data Link layer

[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)] · Hard

Section C — TCP/IP Model

Q74. Which is the standard protocol suite of the Internet?

Internet की मानक protocol suite कौन-सी है?

(A) TCP/IP **(B)** UDP

(C) FTP **(D)** WWW

[Source: IBPS RRB PO Computer Awareness PYQ (via meritnotes) + SSC MTS computer PYQ (via Testbook)] · Easy

Q75. The TCP/IP model was developed _____ the OSI model.

TCP/IP model, OSI model के _____ develop हुआ।

(A) prior to **(B)** after

(C) simultaneous to **(D)** none of these

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q76. The TCP/IP _____ layer is equivalent to the combined Session, Presentation and Application layers of the OSI model.

TCP/IP की _____ layer, OSI model की Session, Presentation और Application layers के समकक्ष है।

(A) application **(B)** network

(C) data link **(D)** physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q77. The number of layers in the Internet protocol stack (reference model) is:

Internet protocol stack (reference model) में layers की संख्या कितनी है?

(A) 5 **(B)** 7

(C) 6 **(D)** 3

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q78. The Data Link layer and Physical layer of the OSI model combine to form the _____ layer of the TCP/IP model.

OSI model की Data Link और Physical layers मिलकर TCP/IP model की _____ layer बनाती हैं।

(A) Network Access **(B)** Internet

(C) Host to Host **(D)** Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Medium

Q79. IP (Internet Protocol) is a protocol of the _____ layer of the TCP/IP model.

IP (Internet Protocol) TCP/IP model की _____ layer का protocol है.

(A) Network Access **(B)** Internet

(C) Transport **(D)** Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Medium

Q80. In TCP/IP, which layer uses specific (user-friendly) addresses — e.g. a URL like example.com?

TCP/IP में कौन-सी layer specific (user-friendly) addresses use करती है — जैसे example.com जैसी URL?

(A) Application layer **(B)** Network layer

(C) Data link layer **(D)** Physical layer

[Source: Forouzan Ch1 (Q1-24) — Practice Set] · Medium

Q81. In the TCP/IP model, a link-layer switch operates at which layers?

TCP/IP model में link-layer switch किन layers पर काम करता है?

(A) Network and Transport layers **(B)** Physical and Data Link layers

(C) Application layer only **(D)** All five layers

[Source: Forouzan Ch1 (Q1-16) — Practice Set] · Medium

Q82. The TCP/IP model is also known as:

TCP/IP model को और क्या कहा जाता है?

(A) OSI model **(B)** DoD model

(C) ISO model **(D)** IEEE model

[Source: Practice — original (textbook-based)] · Medium

Q83. Which layer of the TCP/IP model has the protocols IP, ARP and ICMP?

TCP/IP model की किस layer में IP, ARP, ICMP protocols hote hain?

(A) Application **(B)** Transport

(C) Internet **(D)** Network Access

[Source: Practice — original (textbook-based)] · Medium

Q84. Which TCP/IP layer combines the functions of OSI's session, presentation and application layers?

TCP/IP की कौन-सी layer OSI की session, presentation और application का काम करती है?

(A) Transport **(B)** Internet
(C) Application **(D)** Network Access

[Source: Practice — original (textbook-based)] · Medium

Q85. In the TCP/IP model, process-to-process (program-to-program) communication is provided by the:

TCP/IP model में process-to-process (program-to-program) संवाद किसके द्वारा होता है?

(A) Internet layer **(B)** Transport layer
(C) Network access layer **(D)** Application layer

[Source: Practice — original (textbook-based)] · Medium

Q86. TCP/IP model does not have _____ layer but the OSI model has it.

TCP/IP model में _____ layer नहीं है पर OSI model में है।

(A) Session layer **(B)** Presentation layer
(C) Application layer **(D)** Both (a) and (b)

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

Q87. Which of the following is NOT a part of the TCP/IP model?

निम्नलिखित में से कौन TCP/IP model का हिस्सा नहीं है?

(A) Application **(B)** Transport
(C) Data link **(D)** Session

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Hard

Q88. How many layers are there in the TCP/IP model (original 4-layer suite)?

TCP/IP model (मूल 4-layer suite) में कितनी layers हैं?

(A) Seven **(B)** Five
(C) Four **(D)** Nine

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Hard

Q89. Which of the following is NOT a layer of the TCP/IP model? (Host-to-Host / Internet / Network Access / Physical)

निम्नलिखित में से कौन TCP/IP model की layer नहीं है? (Host-to-Host / Internet / Network Access / Physical)

(A) Host to Host **(B)** Internet
(C) Network Access **(D)** Physical

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Hard

Section D — Comparison, Encapsulation & PDUs

Q90. Identify the correct order of data encapsulation from Layer 7 to Layer 1:

Layer 7 से Layer 1 तक data encapsulation का सही क्रम पहचानिए:

(A) Data → Segment → Packet → Frame → Bits **(B)** Data → Packet → Segment → Frame → Bits
(C) Bits → Frame → Packet → Segment → Data **(D)** Data → Frame → Packet → Segment → Bits

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q91. Encapsulation happens from:

Encapsulation किस दिशा में होती है?

(A) Top to bottom **(B)** Bottom to top
(C) Random **(D)** Parallel

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q92. In the layer hierarchy, as the data packet moves from the upper to the lower layers, headers are:

Layer hierarchy में, जब data packet ऊपर से नीचे की layers की ओर जाता है, तो headers:

(A) Added **(B)** Removed
(C) Rearranged **(D)** Modified

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q93. In TCP/IP, which data unit is encapsulated INSIDE a frame?

TCP/IP में कौन-सा data unit frame के अंदर encapsulate होता है?

(A) A segment **(B)** A datagram
(C) A bit **(D)** A message

[Source: Forouzan Ch1 (Q1-20) — Practice Set] · Medium

Q94. In the TCP/IP model, the unit of data at the transport layer is called a:

TCP/IP model में transport layer पर data की unit कहलाती है:

(A) Packet **(B)** Segment
(C) Frame **(D)** Stream

[Source: Forouzan Ch1 (Q1-20/21) — Practice Set] · Medium

Q95. A system uses five protocol layers. The application creates a 100-byte message and EVERY layer (including the 5th and the 1st) adds a 10-byte header. What is the efficiency (application bytes / total transmitted bytes)?

एक system में पाँच protocol layers हैं। Application 100-byte message बनाती है और हर layer (5वीं और 1वीं सहित) 10-byte header जोड़ती है। Efficiency कितनी होगी (application bytes / कुल transmitted bytes)?

(A) 50% **(B)** 66.7%
(C) 75% **(D)** 80%

[Source: Forouzan Ch1 (P1-11) — Practice Set (Problem)] · Medium

Q96. At the receiver, the correct order of removing overheads is:

Receiver पर overheads हटाने का सही क्रम है:

(A) Physical → Data Link → Network → Transport **(B)** Transport → Network → Data Link → Physical
(C) Network → Physical → Transport → Data Link **(D)** Application → Network → Physical

[Source: Practice — original (textbook-based)] · Medium

Q97. Which combination of address and PDU is CORRECT?

Address और PDU का कौन-सा युग्म सही है?

(A) Data Link — frame — IP address **(B)** Network — packet — MAC address
(C) Transport — segment — port number **(D)** Physical — frame — port number

[Source: Practice — original (textbook-based)] · Medium

Q98. On the sending host, encapsulation adds headers in which order as the message moves down the OSI stack?

Sending host पर message नीचे OSI stack में जाते समय headers किस क्रम में जुड़ते हैं?

(A) Application → Transport → Network → Data Link
(B) Data Link → Network → Transport → Application
(C) Network → Application → Transport → Data Link
(D) Transport → Application → Data Link → Network

[Source: Practice — original (textbook-based)] · Medium

Q99. A source S and destination D are connected through ONE intermediate router R. How many times does a packet visit the Network layer and the Data Link layer during the journey?

Source S और destination D के बीच एक intermediate router R है। यात्रा के दौरान packet Network layer और Data Link layer को कितनी बार visit करता है?

(A) Network layer - 4 times, Data link layer - 4 times
(B) Network layer - 4 times, Data link layer - 6 times
(C) Network layer - 2 times, Data link layer - 4 times
(D) Network layer - 3 times, Data link layer - 4 times

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q100. The process of adding headers (and a trailer) as data moves down the layers is called:

Layers से नीचे जाते समय headers जोड़ने की प्रक्रिया है:

(A) De-encapsulation **(B)** Encapsulation
(C) Fragmentation **(D)** Modulation

[Source: Practice — original (textbook-based)] · Hard

Answer Key — sirf letters (no solutions)

Q1: B **Q2:** A **Q3:** C **Q4:** C **Q5:** D **Q6:** A **Q7:** A **Q8:** A **Q9:** C **Q10:** B
Q11: B **Q12:** B **Q13:** B **Q14:** A **Q15:** A **Q16:** B **Q17:** B **Q18:** A **Q19:** A **Q20:** A
Q21: C **Q22:** A **Q23:** B **Q24:** B **Q25:** B **Q26:** B **Q27:** B **Q28:** C **Q29:** C **Q30:** A
Q31: C **Q32:** B **Q33:** B **Q34:** A **Q35:** B **Q36:** C **Q37:** A **Q38:** D **Q39:** B **Q40:** C
Q41: A **Q42:** B **Q43:** A **Q44:** C **Q45:** D **Q46:** C **Q47:** B **Q48:** B **Q49:** B **Q50:** B
Q51: A **Q52:** B **Q53:** B **Q54:** A **Q55:** A **Q56:** A **Q57:** C **Q58:** B **Q59:** B **Q60:** B
Q61: A **Q62:** C **Q63:** C **Q64:** A **Q65:** A **Q66:** A **Q67:** D **Q68:** D **Q69:** D **Q70:** D
Q71: B **Q72:** D **Q73:** C **Q74:** A **Q75:** A **Q76:** A **Q77:** A **Q78:** A **Q79:** B **Q80:** A
Q81: B **Q82:** B **Q83:** C **Q84:** C **Q85:** B **Q86:** D **Q87:** D **Q88:** C **Q89:** D **Q90:** A
Q91: A **Q92:** A **Q93:** B **Q94:** B **Q95:** B **Q96:** A **Q97:** C **Q98:** A **Q99:** D **Q100:** B